

Asymmetric Attractor Structures as a Source of Persistent Matter–Antimatter Imbalance

Abstract

Recent work has established that discrete, symmetric dynamical systems may exhibit intrinsically asymmetric attractor structures without explicit symmetry breaking. In this article, we explore a possible physical interpretation of such structures in the context of matter–antimatter imbalance. Without introducing new microscopic dynamics, we argue that persistent asymmetries may arise from stability properties of the underlying state space itself. The discussion is deliberately interpretative and does not modify or replace standard baryogenesis mechanisms.

1 Introduction

One of the most persistent open problems in fundamental physics is the observed imbalance between matter and antimatter. Standard quantum field theory permits the creation of matter and antimatter in symmetric pairs, yet the observable universe is overwhelmingly matter-dominated.

Traditional explanations rely on dynamical symmetry-breaking mechanisms, often summarized by the Sakharov conditions. In contrast, the present work explores a complementary viewpoint: that long-lived asymmetries may originate from the *stability structure* of the state space itself rather than from explicit microscopic violations.

This article does not propose a new particle-physics model. Instead, it offers an interpretation of recently established mathematical results on asymmetric attractors in symmetric systems.

2 Mathematical Background

In a companion paper, a discrete dynamical system was introduced whose evolution is formally symmetric under phase inversion. Despite this symmetry, the system exhibits an asymmetric attractor landscape: certain coherent states possess open basins of attraction, while their symmetry-related counterparts reside on unstable separatrices.

Crucially, the asymmetry arises at the level of invariant measures rather than at the level of the evolution rule itself. The dynamics admits symmetric solutions, but these solutions are dynamically unstable.

3 Interpretative Mapping

We now outline a minimal interpretative correspondence:

- Stable attractors correspond to long-lived, persistent states.
- Unstable separatrices correspond to transient or short-lived states.

- Symmetry-related but dynamically inequivalent states naturally lead to asymmetric occupation probabilities.

Within this interpretation, matter and antimatter need not differ in their fundamental equations of motion. Instead, their effective abundances may differ because one class of states occupies stable regions of the global phase space, while the other does not.

4 Relation to Matter–Antimatter Asymmetry

From this viewpoint, an initially symmetric ensemble evolves toward a distribution dominated by dynamically stable states. Even an infinitesimal bias—arising from fluctuations or initial conditions—becomes amplified by the geometry of the attractor structure.

Importantly, this mechanism does not violate CPT symmetry at the microscopic level. The asymmetry emerges statistically and dynamically, not algebraically.

This perspective does not replace baryogenesis scenarios but may complement them by explaining why extremely small initial asymmetries can persist and become cosmologically relevant.

5 Comparison with Standard Approaches

Conventional approaches attribute matter–antimatter imbalance to explicit CP-violating interactions combined with out-of-equilibrium dynamics. The attractor-based interpretation shifts emphasis from interaction terms to global state-space structure.

Both approaches may coexist. In particular, microscopic CP violation may provide the initial perturbation, while attractor asymmetry determines its long-term survival.

6 Limitations

The present discussion is intentionally conservative. No claim is made that the described mechanism quantitatively accounts for the observed baryon asymmetry. No new experimental signatures are proposed.

The interpretation relies on analogy rather than derivation, and further work would be required to embed such attractor structures into concrete quantum or statistical field-theoretic frameworks.

7 Conclusion

We have outlined a possible interpretative bridge between asymmetric attractor structures in symmetric dynamical systems and the persistence of matter over antimatter. The key insight is that asymmetry may arise from stability properties of state space rather than from explicit symmetry breaking.

This viewpoint suggests that the geometry of dynamical systems may play a more fundamental role in cosmological asymmetries than is usually assumed.